Zonghe Shao

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Education

Huazhong University of Science & Technology (HUST) *Sep.2020-Jul.2024*
B.Eng. in ***Artificial Intelligence***

- Overall GPA: 3.93/4.0 (92.0/100) rank: 12/100
- National Encouragement Scholarship, Ministry of Education of PRC (**Top** 3% Natinal Wide), **TWICE**

Research Interests

GOAL: I hope to build a perceptive, interpretable and general artificial intelligence.
Machine Learning & Computer Vision

- I conducted research on time-series prediction and analysis, and now I work on improving the accuracy and robustness of the algorithm
- I am interested in applying computer vision technology to medical treatments, particularly in analyzing human facial expressions and body movements, and I am also intrigued by medical image processing
- I am eager to extract depth information from images and reconstruct the 3D world, with a specific emphasis on cameras and autonomous driving.
- I am also fascinated by deep reinforcement learning, specifically in the area of game decision-making with multiple agents or optimization theory

Publications

A Novel Data-Driven LSTM-SAF Model for Power System Transient Stability Assessment
Zonghe Shao, Qichao Wang, Yuzhe Cao, Defu Cai, Renzhi Lu
IEEE Transactions on Industrial Informatics 2023 (In Submission)

Research Experience

Hao Lu's Research Group, HUST Wuhan, China
Advisor: Assistant Prof. Hao Lu *Mar.2023-Jun.2023*
Topic: Defocus Blur Detection

- Reproduced the mainstream semantic segmentation model on the DUT-DBD dataset, CUHK dataset and analyzed the difficulty of the application to the topic
- Designed a defocus blur detection algorithm that incorporates multi-scale features and partial semantic information
- **Todo**: Make the algorithm as accurate as possible to distinguish between defocus and semantics, while improving the performance of the algorithm in detailed region detection

HAIV Lab, HUST Wuhan, China
Advisor: Assistant Prof. Xiang Xiang *Dec.2022-Present*
Topic: Patient Pain Intensity Estimation Based on 3D Face in Complex Scenarios

- Studied the robust representation of human face based on action units and learned the method for recognizing asymmetric faces in harsh environments
- Constructed a heat map-based intensity analysis model for face action units, the algorithm was applied to the problem of patient pain intensity estimation
- **Todo**: benchmark the intensity of action units, propose facial treatment strategy to improve model robustness, learn the 3D face generation and reconstruction

Deep Reinforcement Learning Lab, HUST Wuhan, China
Advisor: Assistant Prof. Renzhi Lu *Feb.2022-Jan.2023*
Topic: Time-series Data Analysis Based on Machine Learning

- Proposed efficient feature extraction method based on power angle clusters and improved wrapper for power system stability assessment with variable topology

- Redesigned the focal loss function and made it change the parameters to focus on a small number of samples during training
- Constructed an LSTM-SAF method that adequately mine the information albeit in small datasets and gave validation in multiple datasets

Project Experience

Arena Challenge Project *Apr.2023-May.2023*
Primary Developer

- Designed a control system based on improved DWA algorithm in MATLAB to control the movement of the smart car in unknown scenario using only the local area information scanned
- Made the smart car avoid all static and moving obstacles in the scene and achieved a perfect score on all difficulty maps in the challenge

Artificial intelligence elite training program in Baidu *Nov.2022-Present*
Remote Member

- Learned and mastered the design and application of AI algorithms based on PaddlePaddle in Baidu
- Provided standard solutions to open algorithm design problems on the platform matiji

Design of the software named "Ping An Good Car Owner" *Dec.2020-May.2021*
Team Leader

- Constructed a human-computer interaction software based on C language and designed a data management method based on the dual mode of user and administrator
- Realized a variety of software functions, such as vehicle registration, insurance purchase and claim settlement

Extra-curriculum Activity

Alumni Association and Student Development Guidance Center *Oct.2021-Oct.2022*
Member

- Planned and organized several alumni homecoming lectures and acted as the venue director
- Arranged and coordinated the schedule of returning alumni in the 70th anniversary of the university and hosted the Light Run on campus with over 10,000 participants

Grants Committee *Oct.2020-Sep.2022*
Director

- Manage and review students' qualifications for financial aid and scholarships
- Hosted public campaign speeches for various scholarships and honors and led the Grants Committee to campus team of the year honors during my tenure as head

Volunteer Association *Sep.2020-Present*
Member

- Persisted in volunteering activities and was determined to contribute my efforts in environmental protection and social service.
- Served as a session leader and assisted the moderator in organizing the session agenda at the AICOOV

Awards and Honors

Scholarship of National Encouragement, TWICE *Sep.2022 & Sep.2021*
Scholarship of Academic Excellence, TWICE *Sep.2022 & Sep.2021*
Meritorious Winner, Mathematical Contest in Modeling (MCM) *Feb.2022*
National Second Prize, The Chinese Mathematics Competition (CMC) *Nov.2021*
Scholarship of Outstanding Freshman *Jan.2021*

Skills and Interests

- **Languages**: Chinese(native), English (fluent, **academic writing**)
- **Programming**: Python, MATLAB, C/C++
- **DL Framework**: Pytorch, Tensorflow
- **Amateur Interests**: table tennis, badminton, yoga